



python threading to turn on and off LED



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## ✦ AI Overview



Python's `threading` module can be utilized to control an LED (e.g., connected to a Raspberry Pi's GPIO pins) by turning it on and off in a separate thread, allowing the main program to continue its execution without being blocked by `time.sleep()` calls.

Here's a basic approach:

### Import necessary modules:

`threading` for thread management and `time` for delays. You'll also need a library to interact with GPIO pins, such as `gpiozero` or `RPi.GPIO`, depending on your setup.

### Define a function for LED control:

This function will contain the logic to turn the LED on and off. It should take arguments for the LED object and potentially a `threading.Event` object to signal when to stop.

Python



```
import threading
import time
from gpiozero import LED # Or RPi.GPIO
```

```
:param led_object: The LED object (e.g., from gpiozero).
:param stop_event: A threading.Event to signal the thread
:param on_time: Time in seconds for the LED to be on.
:param off_time: Time in seconds for the LED to be off.
"""

while not stop_event.is_set():
    led_object.on()
    time.sleep(on_time)
    led_object.off()
    time.sleep(off_time)
led_object.off() # Ensure LED is off when the thread stops
```

- **Create and start the thread:** In your main program, instantiate the LED object, create a `threading.Event` object, and then create and start the `threading.Thread`, passing the LED control function and its arguments.

Python



```
# Initialize the LED (e.g., connected to GPIO pin 17)
led = LED(17)

# Create an Event object to control the thread
stop_blinking_event = threading.Event()

# Create and start the thread
```

- **Control the thread:** In your main program, you can perform other tasks. When you want to stop the LED blinking, set the `stop_event` and `join()` the thread to wait for it to finish.

Python



```
# Main program continues execution
print("LED blinking in the background...")
time.sleep(5) # Simulate other tasks

# Stop the blinking
print("Stopping LED blinking...")
stop_blinking_event.set()
led_thread.join()
print("LED blinking stopped.")
```

This approach allows for independent control of the LED, preventing the main program from being blocked while the LED blinks or remains in a specific state.

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## Python threading class for GPIO Led blink

I want to keep blinking **led** asynchronous (obviously) and have the control on `Led16()` object status, maybe without creating new threads each time I call its ...